NSE Equity Intelligence Platform — Master Consolidated Specification (v6 Final)

1. Executive Architecture & System Design

The NSE Equity Intelligence Platform is an institutional-grade quantitative research and equity analysis engine for Indian National Stock Exchange (NSE) equities. The platform enforces strict point-in-time correctness, mathematical consistency across corporate actions, dual-era Indian taxation modeling, and complete separation between raw facts, derived metrics, and forecast models.

2. Point-in-Time Data Semantics & Schema

To eliminate lookahead bias and survivorship bias:

1. **Three-Timestamp Temporal Tracking**:
    - `published_at`: When the company/exchange officially disclosed the information.
    - `ingested_at`: When our ingestion pipeline downloaded, hashed, and committed the data.
    - `restated_at`: When an amended filing or correction was published (NULL for originals).
2. **Point-in-Time Rule**: Backtests and historical queries gate strictly on `ingested_at <= :as_of_timestamp`. Even if a company published at 19:00 on Day T, a model querying as of 18:00 on Day T cannot access it.
3. **Restatement Versioning**: Restatements do not overwrite previous filings. The previous record has its `valid_to` set to the restatement's `ingested_at`, while the new record has `valid_from = restatement.ingested_at`.
4. **Flexible EAV Financials**: Avoids hardcoded statement columns. Supports standard non-financials, banks (NIM, CASA, GNPA), and insurance companies with explicit units (`unit_multiplier`) and reporting level (`consolidated`).

3. Corporate Actions & Total Return Methodology

3.1 Adjustment Factor Definitions

All corporate actions express ratios unambiguously as `new_shares : old_shares`.

- **Split (n: 1n : 1)**: $f = 1/n$f = 1/n.
- **Bonus (b: eb : e)**: $f = e/(e+b)$f = e/(e + b).
- **Rights Issue**:

  $$\text{TERP} = \frac{P_{\text{cum}} \cdot n_{\text{old}} + P_{\text{sub}} \cdot n_{\text{new}}}{n_{\text{old}} + n_{\text{new}}}$$

  TERP = nold + nnewPcum · nold + Psub · nnew

  $$f_{\text{rights}} = \frac{\text{TERP}}{P_{\text{cum}}}$$

  frights = PcumTERP
- **Cumulative Factor** for trade date tt:

  $$F(t) = \prod_{i:\, d_i > t,\, \text{action}_i \in \{\text{split, bonus, rights}\}} f_i$$

  F(t) = i:di>t,actioni∈{split, bonus, rights}∏ fi

  $$\text{close_adj}(t) = \text{close_raw}(t) \times F(t)$$

  close_adj(t) = close_raw(t) × F(t)

  $$\text{volume_adj}(t) = \text{volume_raw}(t) \div F(t)$$

  volume_adj(t) = volume_raw(t) ÷ F(t)

3.2 Total Return Index (TRTR) with Dividend Reinvestment

Dividends are stored strictly as reported (D_{raw} Draw). The dividend per share is adjusted to the historical share basis:

$$D_{\text{adj}}(d) = D_{\text{raw}}(d) \times F(d)$$

Dadj(d) = Draw(d) × F(d)

Reinvestment convention: Cash dividends are reinvested on the ex-dividend date at the ex-date adjusted closing price $\text{close_adj}(d)$close_adj(d).

$$\text{TR}(d) = \text{TR}(d-1) \times \frac{\text{close_adj}(d) + D_{\text{adj}}(d)}{\text{close_adj}(d-1)}$$

TR(d) = TR(d − 1) × close_adj(d − 1)close_adj(d) + Dadj(d)On non-dividend dates:

$$\text{TR}(d) = \text{TR}(d-1) \times \frac{\text{close_adj}(d)}{\text{close_adj}(d-1)}$$

TR(d) = TR(d − 1) × close_adj(d − 1)close_adj(d)

4. Dual-Era Indian Taxation Engine

The system bridges the historic transition from the Income-tax Act, 1961 to the Income-tax Act, 2025:

Category	Legacy Law (< 2026-04-01)	Modern Law (>= 2026-04-01)	Status	Official Source Reference
Governing Statute	Income-tax Act, 1961	Income-tax Act, 2025	✅ Verified	Ministry of Law & Justice, Gazettes
Period Identifier	Financial Year (FY) / Assessment Year (AY)	**Tax Year** (e.g., TY 2026-27)	✅ Verified	IT Act 2025 Sec 3
STCG on Listed Equity (STT paid)	Sec 111A: 15% (< 2024-07-23), 20% (>= 2024-07-23)	**Sec 196**: 20%	✅ Verified	Finance (No. 2) Act 2024 / IT Act 2025
LTCG on Listed Equity	Sec 112A: 10% (> ₹1L up to 2024-07-22), 12.5% (> ₹1.25L from 2024-07-23)	**Sec 198**: 12.5% above ₹1.25 Lakh / TY	✅ Verified	Finance (No. 2) Act 2024 / IT Act 2025
Dividend TDS (Resident, with PAN)	Sec 194: 10% (threshold ₹5k up to FY24-25, ₹10k in FY25-26)	**Sec 393(1)**: 10% (threshold ₹10,000 / company / TY)	✅ Verified	Finance Act 2020/2025 / IT Act 2025
Dividend TDS (No PAN)	Sec 206AA: 20%	Sec 393 / Sec 206AA equivalent: 20%	✅ Verified	IT Act Sec 206AA
Loss Set-off (Intra-head)	Sec 70: STCL offsets STCG/LTCG; LTCL offsets only LTCG	**Sec 108**: STCL offsets STCG/LTCG; LTCL offsets only LTCG	✅ Verified	IT Act 1961 / IT Act 2025
Loss Carry-forward	Sec 74: 8 assessment years	**Sec 111**: 8 tax years (Sec 536 transitions prior losses)	✅ Verified	IT Act 1961 / IT Act 2025
Company Share Buybacks	Sec 115QA repealed 2024-10-01; taxed as dividend to shareholder	Taxed as dividend in shareholder hands	✅ Verified	Finance (No. 2) Act 2024
STT: Equity Delivery	0.1% on buy AND 0.1% on sell turnover	0.1% on buy AND 0.1% on sell turnover	✅ Verified	Finance (No. 2) Act 2024, 7th Sched.
Stamp Duty	0.015% on buy turnover	0.015% on buy turnover	✅ Verified	Indian Stamp Rules 2020
GST Rate & Base	18% on (Brokerage + Exchange + SEBI + DP); excludes STT & Stamp Duty	Same: SAC 997152, 18% on service charges	✅ Verified	CBIC Notification 11/2017-CT(Rate)
NSE Transaction Charges	0.00345%	0.00345%	⚠️ UNVERIFIED	Circular rate subject to exchange revision

Category	Legacy Law (< 2026-04-01)	Modern Law (>= 2026-04-01)	Status	Official Source Reference
SEBI Turnover Fee	0.0001%	0.0001%	⚠️ UNVERIFIED	SEBI fee schedule subject to annual circular

Safety Guard: Any tax calculation invoking an unverified rate raises `UnverifiedRateError` in production mode unless explicitly executed with `allow_unverified=True`.

5. Verified Financial Formulas & Worked Examples

5.1 DuPont 5-Factor Analysis

$$\text{ROE} = (\frac{\text{Net Income}}{\text{EBT}}) \times (\frac{\text{EBT}}{\text{EBIT}}) \times (\frac{\text{EBIT}}{\text{Revenue}}) \times (\frac{\text{Revenue}}{\text{Assets}}) \times (\frac{\text{Assets}}{\text{Equity}})$$

ROE = (EBTNet Income) × (EBITEBT) × (RevenueEBIT) × (AssetsRevenue) × (EquityAssets)

- **Reference**: Palepu, Healy & Peek, *Business Analysis and Valuation: IFRS Edition*, 6th ed. (2019), Ch. 5.
- **Independent Numerical Test**: Net Income = 120, EBT = 160, EBIT = 200, Revenue = 1000, Assets = 2000, Equity = 800.
  - Tax Burden = $120/160 = 0.7500$120/160 = 0.7500
  - Interest Burden = $160/200 = 0.8000$160/200 = 0.8000
  - EBIT Margin = $200/1000 = 0.2000$200/1000 = 0.2000
  - Asset Turnover = $1000/2000 = 0.5000$1000/2000 = 0.5000
  - Equity Multiplier = $2000/800 = 2.5000$2000/800 = 2.5000
  - Product = $0.75 \times 0.80 \times 0.20 \times 0.50 \times 2.50 = 0.1500$0.75 × 0.80 × 0.20 × 0.50 × 2.50 = 0.1500 (15.00% ROE)
  - Direct ROE = $120/800 = 0.1500$120/800 = 0.1500 (15.00%). Verified identity match.

5.2 Damodaran FCFF Discounted Cash Flow (DCF)

- **Reference**: Damodaran, A., *Investment Valuation*, 3rd ed. (2012), Ch. 15.
- **Independent Numerical Test**: EBIT = 100, Tax = 25%, Depr = 20, CapEx = 30, $\Delta \text{WC} = 5$ΔWC = 5. WACC = 12%, Growth 1-5 = 8%, Terminal $g = 4\%$g = 4%.
  - $\text{FCFF}_0 = 100 \times 0.75 + 20 - 30 - 5 = 60$FCFF0 = 100 × 0.75 + 20 − 30 − 5 = 60.

- PV of 5-year cash flows = $57.86 + 55.79 + 53.80 + 51.87 + 50.02 = 269.34$ 57.86 + 55.79 + 53.80 + 51.87 + 50.02 = 269.34.
- $TV_5 = (60 \times 1.08^5 \times 1.04)/(0.12 - 0.04) = 91.6864/0.08 = 1146.08$TV5 = (60 × 1.085 × 1.04)/(0.12 − 0.04) = 91.6864/0.08 = 1146.08.
- $PV(TV_5) = 1146.08/(1.12^5) = 650.30$PV (TV5) = 1146.08/(1.125) = 650.30.
- Enterprise Value = $269.34 + 650.30 = 919.64$269.34 + 650.30 = 919.64. Verified.

5.3 Reverse DCF Solver

Solves for implied cash flow growth rate g^* g∗ such that $\text{DCF}(g^*) = P_0$ DCF(g∗) = P0.

- **Labeling Rule**: Round-trip passes verify **Numerical Consistency of Solver**, NOT real-world investment correctness. Real-world forecasts carry confidence bands and sensitivity matrices.